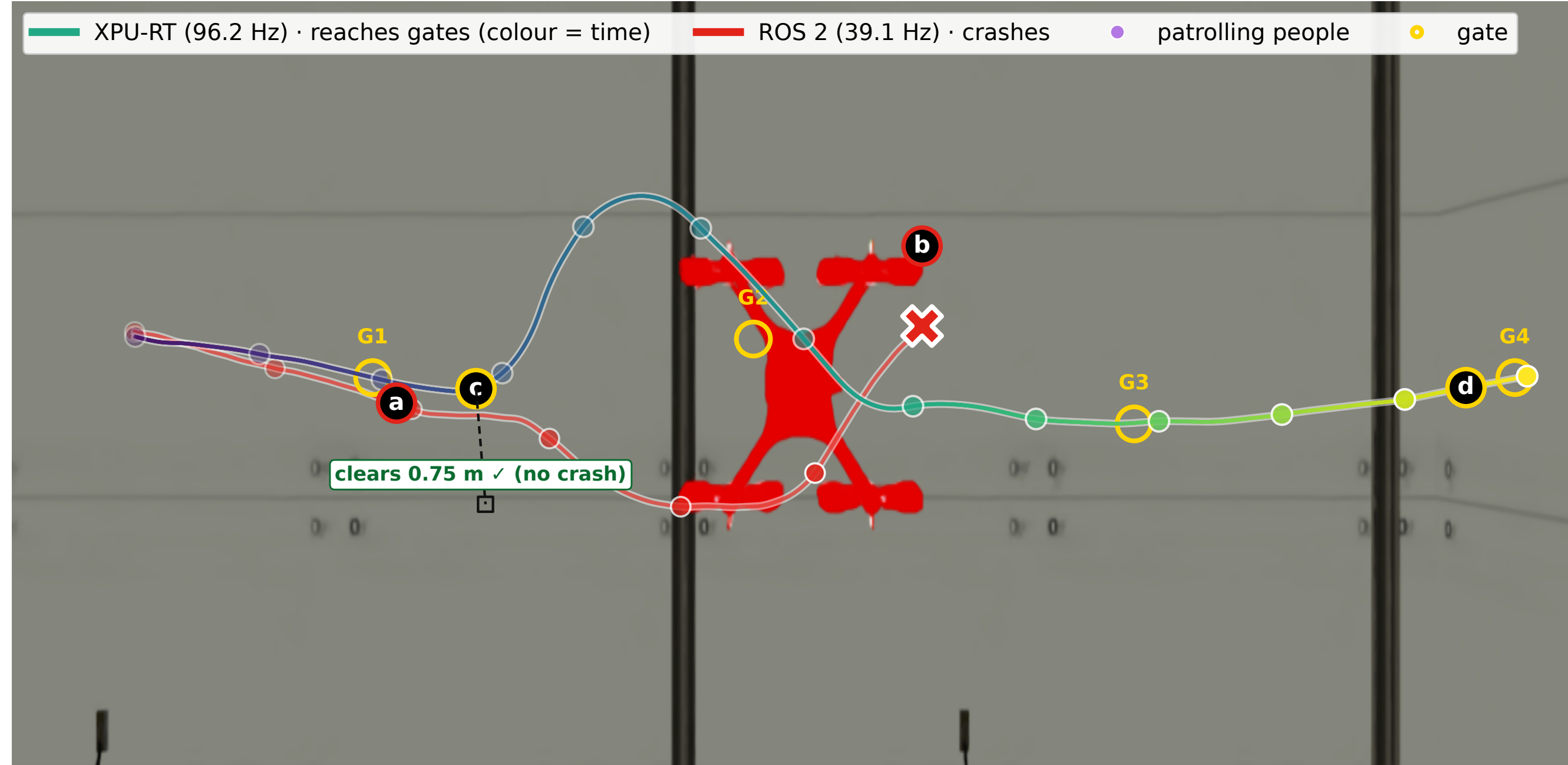
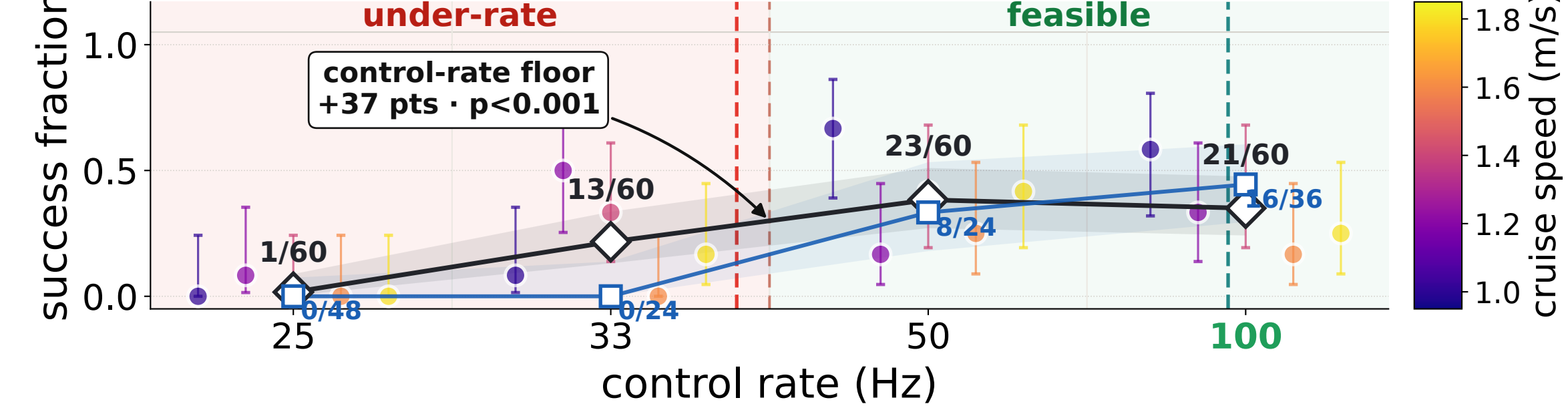


Warehouse gate-course showdown — same aisle, same obstacles, same controller gain
the runtimes differ only in the control cadence measured on the K1: 10.0 ms vs 25.6 ms between control outputs

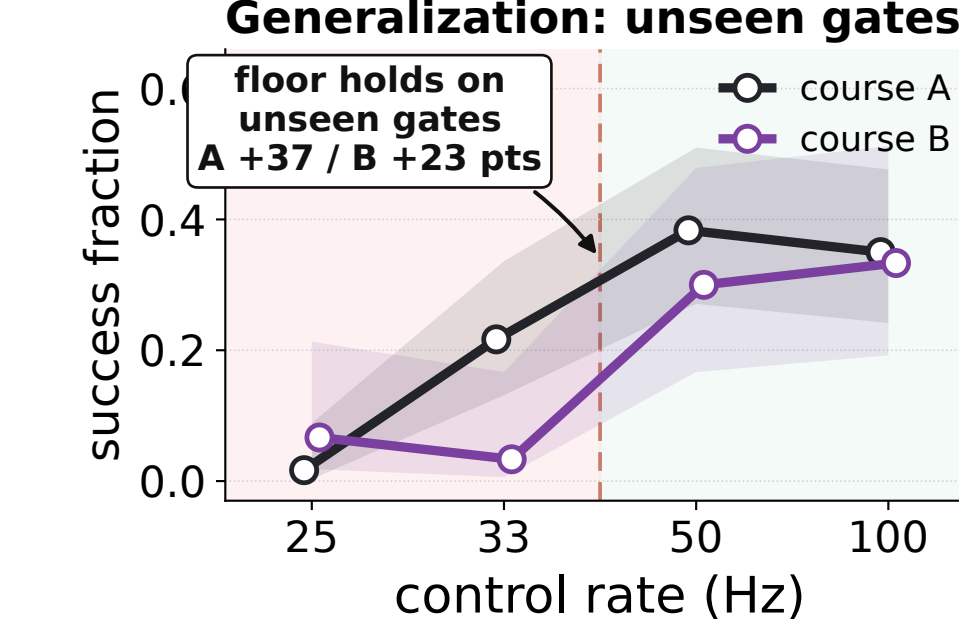
A



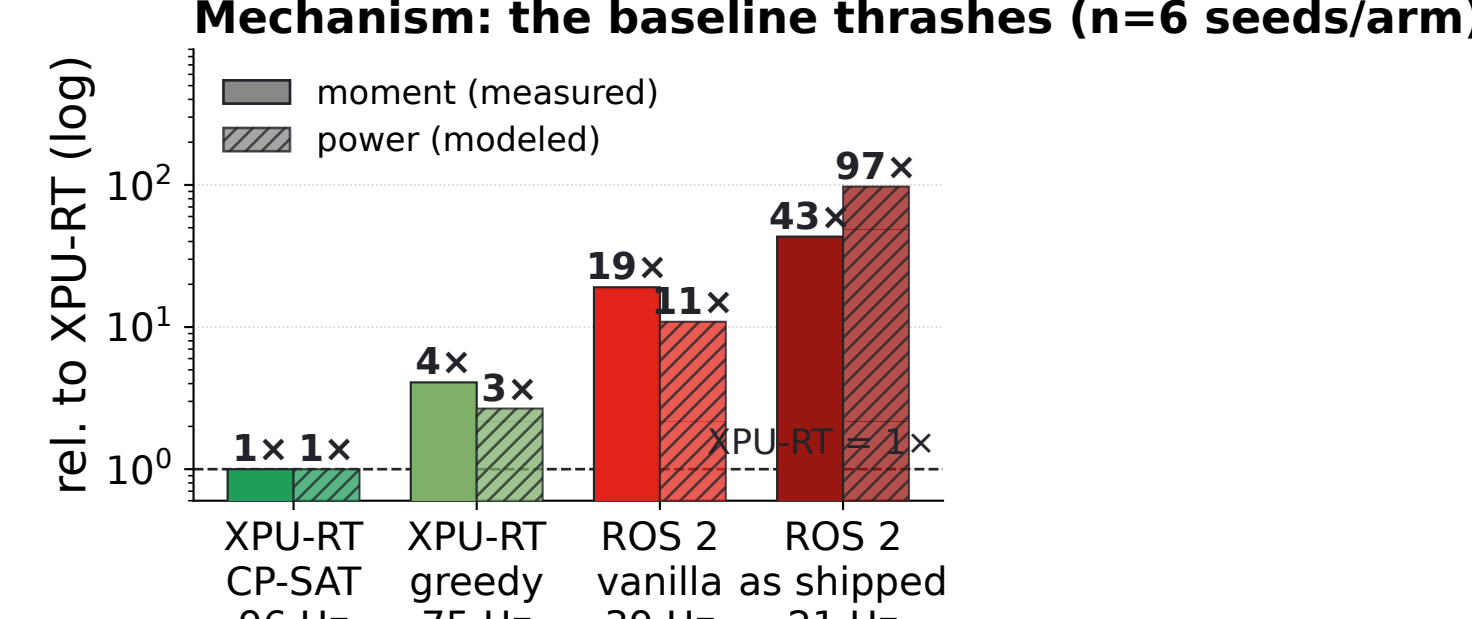
B



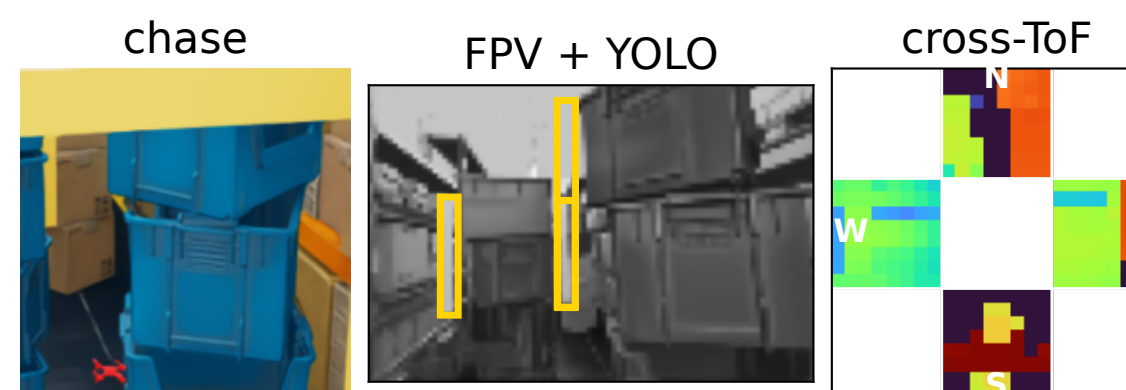
C



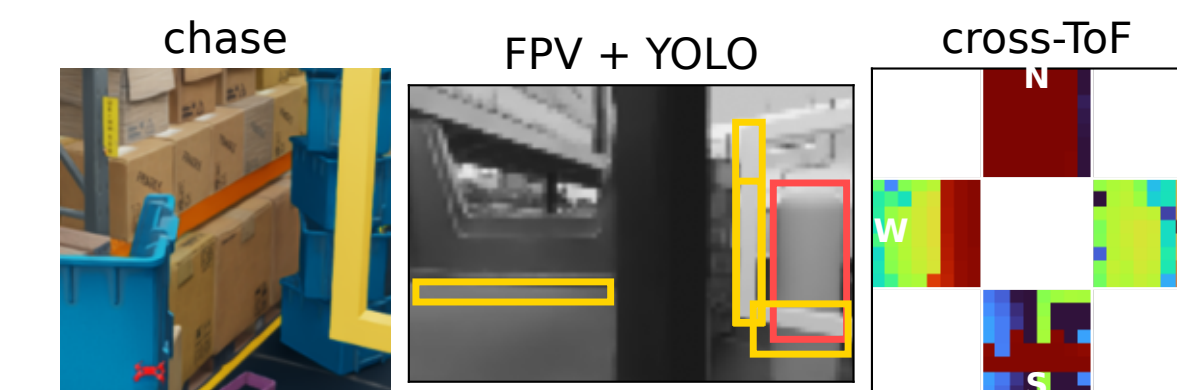
D



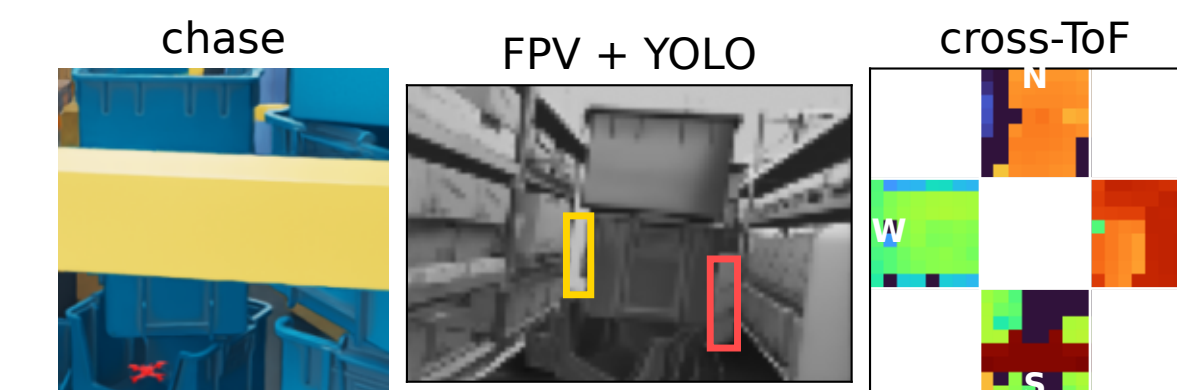
a ROS 2 39 Hz · clears G1 · t=3.6s



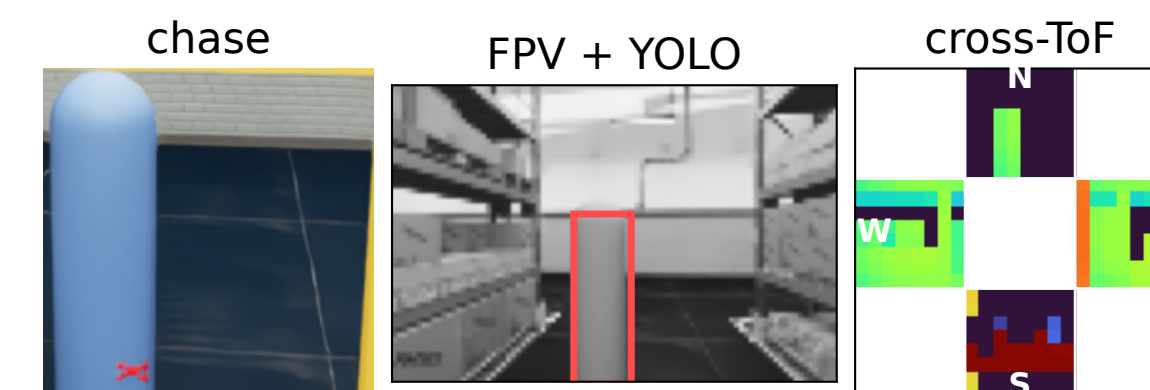
b ROS 2 39 Hz · crash · t=9.5s



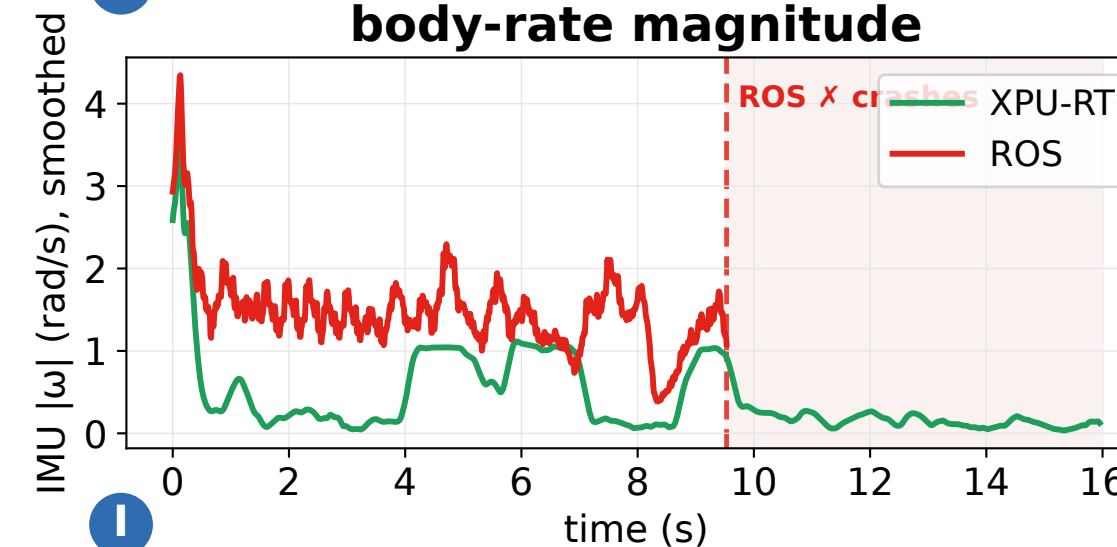
c XPU-RT 96 Hz · clears crate 0.75 m · t=4.3s



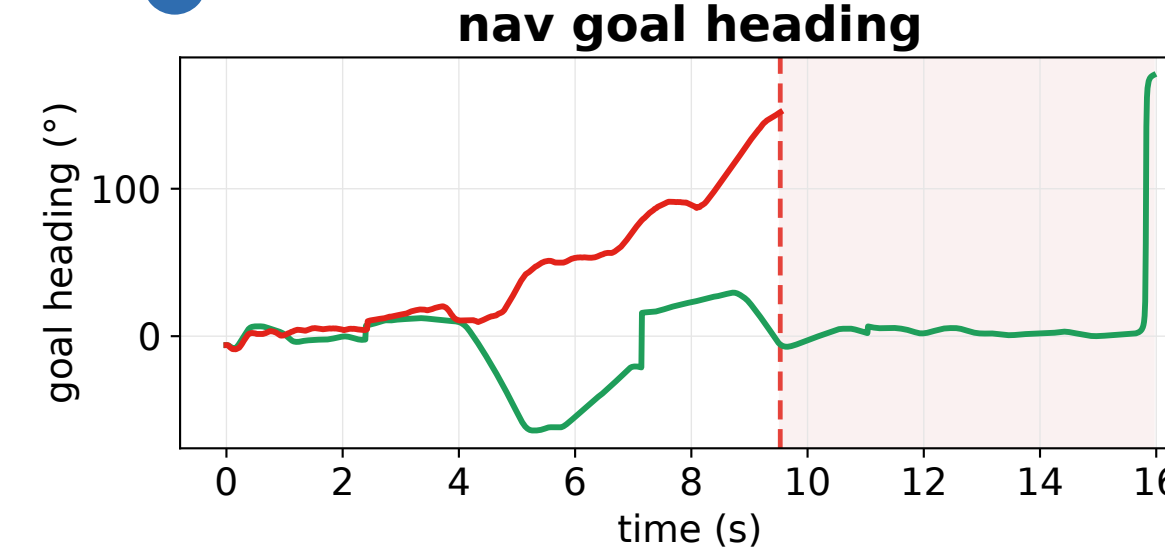
d XPU-RT 96 Hz · reaches gate · t=15.3s



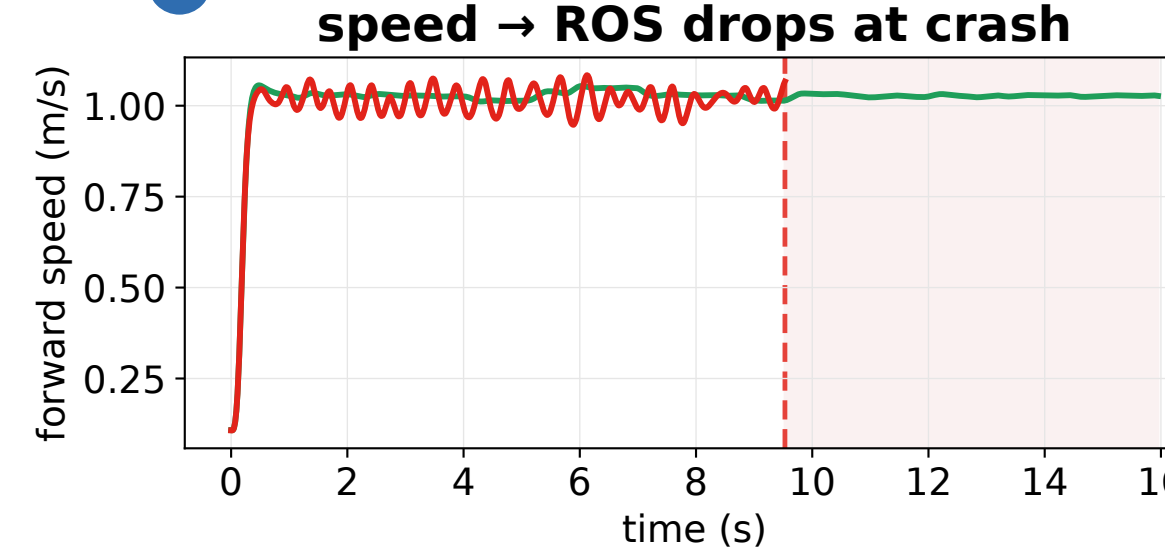
E body-rate magnitude



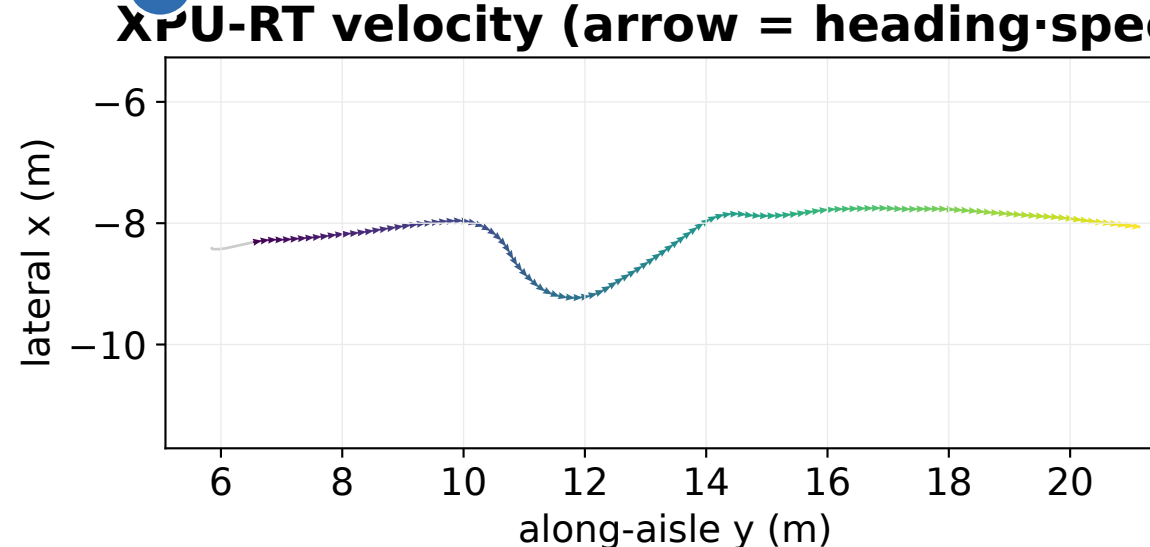
F nav goal heading



G speed → ROS drops at crash



H XPU-RT velocity (arrow = heading-speed)



Onboard K1, measured, 45 Hz camera — XPU-RT-CP-SAT: camera→control 57 ms, every frame on time, control every 10.0 ms · XPU-RT-greedy: camera→control 748 ms, 44 of 44 frames late, control every 13.2 ms · ROS 2 vanilla: camera→control 242 ms, control every 25.6 ms

